

Greenberg Harper 16

Vincent Wang

March 2026

1 Further Applications to Spheres

16.6. Let $f, g : S^n \rightarrow S^n$ st $f(x) \neq g(x)$ for all $x \in S^n$. Then $f \simeq ag$, where a is the antipodal map. Then $H_n(f) = (-1)^{n+1}H_n(g)$.

Proof: Let $F : S^n \times I \rightarrow E^{n+1} - 0$ by $F(x, t) = (1 - t)f(x) - tg(x)$. Since B^{n+1} is convex, we have $F(x, t) \in E^{n+1}$. Further,

$$\begin{aligned} (1 - t)f(x) - tg(x) = 0 &\rightarrow (1 - t)f(x) = tg(x) \rightarrow \|(1 - t)f(x)\| = \|tg(x)\| \\ &\rightarrow 1 - t = t \rightarrow t = 1/2. \end{aligned}$$

Then $f(x) = g(x)$, absurd. Hence, F is well-defined.

Define $H : S^n \times I \rightarrow S^n$ by $H(x, t) = F(x, t)/\|F(x, t)\|$. Note that $H(x, 0) = f(x)/\|f(x)\| = f(x)$ and $H(x, 1) = -g(x)/\|g(x)\| = ag(x)$, so H is a homotopy from f to ag .

$H_n(a) = (-1)^{n+1}$, so $H_n(f) = H_n(ag) = H_n(a)H_n(g) = (-1)^{n+1}H_n(g)$, as desired.

16.7. Let $f : X \rightarrow S^n$ be not onto. Then f is null-homotopic.

Proof: Pick $y_0 \in S^n - f(X)$ and let $g : S^n - y_0 \rightarrow \mathbb{R}^n$ be the stereographic projection. $\mathbb{R}^n$ is contractible, so $gf : X \rightarrow \mathbb{R}^n$ is null-homotopic. Hence, $g^{-1}gf = f : X \rightarrow S^n$ is null-homotopic.

Let $f : S^n \rightarrow S^n$ be null-homotopic. Then f has a fixed point and a point x such that $f(x) = -x$.

Proof: Sps FTSOC $f(x) \neq x$ for all $x \in S^n$. Then $f \simeq a$ by 16.6. Since $f \simeq e_{x_0}$ for some $x_0 \in S^n$, we have $a \simeq e_{x_0}$. Then $H_n(a) = H_n(e_{x_0})$. For $n > 0$, $H_n(e_{x_0}) = 0$ (since the reduced homology of a point is 0), so $(-1)^{n+1}c = 0$ for all $c \in H_n(S^n)$. Since $H_n(S^n) \cong R$ as R -modules, we can pick nonzero $c \in H_n(S^n)$, absurd.

If f is null-homotopic, so is af , so af has a fixed point x , ie $f(x) = -x$ for some $x \in S^n$.

16.8. Any $f : S^{2n} \rightarrow S^{2n}$ has a fixed point or sends some point to its antipode.

Proof: Sps f has no fixed or opposite points. Then $H_{2n}(f) = -1$ and $H_{2n}(af) = -1$. $H_{2n}(a) = (-1)^{2n+1} = -1$, so $\bar{z} = -\bar{z}$ for all $\bar{z} \in H_{2n}(S^{2n})$. Take $R = \mathbb{Z}$. Since $H_{2n}(S^{2n}) \cong \mathbb{Z}$, we have $1 = -1$ in $\mathbb{Z}$, absurd.

16.9. Consider S^3 . We have the maps

$$\begin{aligned}(x_1, x_2, x_3, x_4) &\mapsto (x_2, -x_1, x_4, -x_3) \\ (x_1, x_2, x_3, x_4) &\mapsto (-x_3, x_4, x_1, -x_2) \\ (x_1, x_2, x_3, x_4) &\mapsto (x_4, x_3, -x_2, -x_1)\end{aligned}$$

We can divide the coordinates of S^{4n+3} into blocks of 4 and repeat this construction.

16.11. Let G be a group of homeomorphisms acting freely on S^{2n} (so $gx = x$ for some $x \in S^{2n}$ implies $g = 1$). Then $|G| \leq 2$.

Proof: Sps FTSOC $g \neq 1, a$. Then $g \simeq a$ and $ag \simeq a$, so $g \simeq 1 \simeq a$. Then $H_{2n}(1) = H_{2n}(a)$, so $-1 = 1$ in $\mathbb{Z}$, absurd. Hence, $G = 1, \langle a \rangle$, so $|G| \leq 2$.

16.12. Every $f : P^{2n} \rightarrow P^{2n}$ has a fixed point.

Proof: Let $p : S^{2n} \rightarrow P^{2n}$ be the covering map. Note that $(fp)_*(\pi_1(S^{2n})) = (fp)_*(0) = 0$, S^{2n} is path connected and locally path connected, so by the general lifting theorem, let $g : S^{2n} \rightarrow S^{2n}$ such that $pg = fp$. By 16.8, g has a fixed point or an antipodal point.

Sps g has a fixed point x . Then $p(x) = f(p(x))$, so f has the fixed point $p(x)$. Sps g has an antipodal point x . Then $p(-x) = fp(x)$. By construction, $p(-x) = p(x)$, so f has the fixed point $p(x)$.

2 Projective n -space

Let $P^n = S^n / \{x \sim -x : x \in S^n\}$. Let $p : S^n \rightarrow P^n$ be the quotient map. I claim that p is a covering map. Let $U \subseteq S^n$ be open. Then $p^{-1}p(U) = U \cup a(U)$ by construction (is open, since a is a homeomorphism), so $p(U)$ is open, so p is an open map.

Suppose $A \leq S^n$ such that $\text{diam } A < 2$. Then $A \cap a(A) = \emptyset$.

Proof: Sps FTSOC $x \in A \cap a(A)$, so $x \in A$ and $a(x) \in A$. Then $\text{diam } A \geq \|x - a(x)\| = 2$, absurd.

Pick $U \leq S^n$ with $\text{diam } U < 2$. $U, a(U)$ are disjoint. $p' : U \rightarrow p(U)$ is open injective (since U contains at most one of $x, a(x)$ for all $x \in S^n$) and therefore a homeomorphism. $p'' : a(U) \rightarrow p(U)$ is also a homeomorphism.

Let $\mathcal{U}$ denote the set of all U . $\mathcal{U}$ is a cover of S^n , so $p(\mathcal{U})$ is a cover of P^n . Hence, p is a covering map.

Fix $b_0 \in P^n$ and pick $e_0 \in S^n$ such that $p(e_0) = b_0$. There exists a map $\Phi : \pi_1(P^n, b_0) \rightarrow p^{-1}(b_0) = \{e_0, a(e_0)\}$. Since S^n is path-connected, Φ is surjective. Since S^n is simply connected for $n > 1$, Φ is bijective.*

Sps $[\gamma_1], [\gamma_2] \in \pi_1(P^n, b_0)$ st $\tilde{\gamma}_1(1) = \tilde{\gamma}_2(1)$. Then $\tilde{\gamma}_1 * \overline{\tilde{\gamma}_2}$ is a loop through e_0 . Since $\pi_1(S^n, e_0) = 0$, we have $[\tilde{\gamma}_1 * \overline{\tilde{\gamma}_2}] = [e_{e_0}]$, so $p_*([\tilde{\gamma}_1 * \overline{\tilde{\gamma}_2}]) = [e_{b_0}]$, so

$$p_*([\tilde{\gamma}_1 * \overline{\tilde{\gamma}_2}]) = [p(\tilde{\gamma}_1 * \overline{\tilde{\gamma}_2})] = [p\tilde{\gamma}_1 * \overline{p\tilde{\gamma}_2}] = [\gamma_1 * \overline{\gamma_2}] = [\gamma_1][\gamma_2]^{-1},$$

so $[\gamma_1] = [\gamma_2]$.

The only group of order 2 is $\mathbb{Z}_2$, so $\pi_1(P^n) \cong \mathbb{Z}_2$ for $n > 1$.????????????????????

3 Mayer-Vietoris Sequence

Consider the triple (X, X_1, X_2) where $X_1, X_2 \subseteq X$. The triple is said to be exact if both inclusions

$$\begin{aligned} k_1 : (X_1, X_1 \cap X_2) &\rightarrow (X_1 \cup X_2, X_2) \\ k_2 : (X_2, X_1 \cap X_2) &\rightarrow (X_1 \cup X_2, X_1) \end{aligned}$$

are excisions. Note if either k_1, k_2 is an excision, then so is the other.

Sps X_1, X_2 are open. Note that

$$\overline{X_2 - X_1 \cap X_2} = \overline{X_1 \cup X_2 - X_1} = X_1 \cup X_2 - X_1 \subseteq X_2 = X_2^\circ,$$

so by the Excision theorem, k_1 is an excision.

(S^n, E_n^+, E_n^-) is an exact triad. I claim $k_1 : (E_n^+, S^{n-1}) \rightarrow (S^n, E_n^-)$ is exact. Let $V = \{x \in S^n : \pi_1(x) < -1/2\}$. We have $i : (S^n - V, E_n^- - V) \rightarrow (S^n, E_n^-)$ is an excision by the Excision theorem since $\overline{V} \subseteq (E_n^-)^\circ$. There exists a homotopy

$(E_n^+, S^{n-1}) \rightarrow (S_n - V, E_n^- - V)$, so k_1 is an excision.

Similarly, $k_2 : (E_n^-, S^{n-1}) \rightarrow (S^n, E_n^+)$ is an excision, so the triad is exact.

Let $A, B \subseteq S^n$, $n \geq 2$. Sps closed disjoint A, B do not separate S^n . Then $A \cup B$ does not separate S^n .

Proof: Let $X_1 = S^n - A, X_2 = S^n - B$, so X_1, X_2 are open. Further, $X_1 \cup X_2 = S^n - A \cap B = S^n$. Note that $X_1 \cap X_2 = S^n - A \cup B \neq \emptyset$ since S^n is connected. Then (S^n, X_1, X_2) is an exact triad. Mayer Vietoris gives the exact sequence

$$H_1(S^n) \rightarrow H_0^\#(S^n - A \cup B) \rightarrow H_0^\#(X_1) \oplus H_0^\#(X_2).$$

Since $n \geq 2$, we have $H_1(S^n) = 0$. Since X_1, X_2 are connected, we have the exact sequence $0 \rightarrow H_0^\#(S^n - A \cup B) \rightarrow 0$, ie $S^n - A \cup B$ is connected.

If A, B are connected open and $A \cup B = S^n$, then $A \cap B$ is connected.

Proof: Let $A' = S^n - A, B' = S^n - B$. Then A', B' do not separate S^n , $A' \cap B' = S^n - A \cup B = \emptyset$, and are closed. By the previous part, $A' \cup B'$ does not separate S^n , ie $S^n - A' \cup B' = A \cap B$ is connected.

4 Jordan Separation Theorem

Let $f : S^2 \rightarrow \mathbb{R}^2$ be continuous. I claim there exists a point $x_0 \in S^2$ such that $f(x_0) = f(-x_0)$.

Proof: Suppose otherwise. Let $g : S^2 \rightarrow S^1$ be given by

$$g(x) = \frac{f(-x) - f(x)}{\|f(-x) - f(x)\|},$$

so g is continuous antipode preserving. Consider the inclusion $i : S^1 \rightarrow S^2$ given by $i(x_1, x_2) = (x_1, x_2, 0)$. Thus, we have a continuous antipode preserving map $fi : S^1 \rightarrow S^1$.

Lemma: Let $f : S^1 \rightarrow S^1$ be continuous antipode preserving. Then f is not null-homotopic.

Proof: Let $p : \mathbb{R} \rightarrow S^1$ be the standard covering map. Let $p' : I \rightarrow S^1$ be a restriction of p . Consider $\gamma = fp'$, a loop in S^1 , say about b_0 . Let $p(e_0) = b_0$ for $e_0 \in \mathbb{R}$. Note for $s \in [0, 1/2]$,

$$\gamma(s + 1/2) = fp'(s + 1/2) = f(-p'(s)) = -fp(s) = -\gamma(s).$$

Let $\tilde{\gamma}$ be the lift of γ at e_0 . Then

$$p\tilde{\gamma}(s + 1/2) = -p\tilde{\gamma}(s) = p(\tilde{\gamma}(s) + 1/2),$$

so $\gamma(s + 1/2) = \gamma(s) + 1/2 + k_s$ for some $k_s \in \mathbb{Z}$. Since $\tilde{\gamma}$ is continuous, $k : [0, 1/2] \rightarrow \mathbb{R}$ given by $k(s) = \gamma(s + 1/2) - \gamma(s) - 1/2$ is continuous. Since $\text{im } k \subseteq \mathbb{Z}$, we have k is constant.

Then

$$\tilde{\gamma}(1) = \tilde{\gamma}(1/2) + 1/2 + k = \tilde{\gamma}(0) + 1 + 2k.$$

Since $2k + 1 \neq 0$ since $k \in \mathbb{Z}$, we have $\tilde{\gamma}(1) \neq \tilde{\gamma}(0)$, ie γ is not nullhomotopic. Since γ is not nullhomotopic, neither is f .

Now $fi : S^1 \rightarrow S^1$ is not nullhomotopic. However, by the identification $E_2^+ \cong B^2$, fi extends to $h : B^2 \rightarrow S^1$, so fi is null-homotopic, absurd.

Let $s_r < S^n$ denote a homeomorph of S^r for $r < n$. I claim that

$$H_q^\#(S^n - s_r) = \begin{cases} R & \text{if } q = n - r - 1, \\ 0 & \text{if } q \neq n - r - 1. \end{cases}$$

First, I claim that indeed $s_r \neq S^n$. If $s_r \rightarrow S^n$ is a homeomorphism, then $H_r(s_r) = H_r(S^n) \rightarrow R = 0$, absurd. I induct on r . For $r = 0$, $S^n - s_r \simeq \mathbb{R}^n - \{0\} \simeq S^{n-1}$, so $H_q^\#(S^n - s_0) = R$ if $q = n - 1$ and 0 otherwise. Assume the result for $r - 1$.

Let $\phi : S^r \rightarrow S^n$ be the inclusion. Denote by $e_r^+ = \phi(E_r^+)$ and $e_r^- = \phi(E_r^-)$. Let $X_1 = S^n - e_r^+$ and $X_2 = S^n - e_r^-$. Note that $X = X_1 \cup X_2 = S^n - e_r^+ \cap e_r^- = S^n - s_{r-1}$, where $e_{r-1} = \phi(S^{r-1})$, where we have the inclusion $S^{r-1} \hookrightarrow S^r$ and $A = X_1 \cap X_2 = S^n - e_r^+ \cup e_r^- = S^n - s_r$. X_1, X_2 are open, so (X, X_1, X_2) is an exact triad.

Note that $A \neq \emptyset$, so we can apply Mayer Vietoris with reduced homology. We have the exact sequence

$$H_{q+1}(X_1) \oplus H_{q+1}(X_2) \rightarrow H_{q+1}(X) \rightarrow H_q^\#(A) \rightarrow H_q^\#(X_1) \oplus H_q^\#(X_2).$$

Since e_r^+, e_r^- are homeomorphs of e^r , we have that $H_{q+1}(X_1), H_{q+1}(X_2), H_q^\#(X_1), H_q^\#(X_2)$ vanish, so $H_{q+1}(X) \cong H_q^\#(A)$. Then

$$H_q^\#(S^n - s_r) = \begin{cases} R & \text{if } q + 1 = n - (r - 1) - 1, \\ 0 & \text{otherwise,} \end{cases}$$

as desired.

Sps FTSOC $s_n \neq S^n$. Then $H_{-1}^\#(S^n - s_n) \cong H_0^\#(S^n - s_{n-1}) \cong R$, absurd.

Let $f, g : S^n \rightarrow S^n$ be continuous. If $|\deg f| \neq |\deg g|$, then $f(x) \cdot g(x) = 0$ for some $x \in S^n$.

Proof: Sps $f(x) \cdot g(x) \neq 0$ for all $x \in S^n$. The map $x \mapsto f(x) \cdot g(x)$ is continuous. If $f(x_1) \cdot g(x_1) < 0 < f(x_2) \cdot g(x_2)$ for some $x_1, x_2 \in S^n$, since S^n is connected (for $n \geq 1$), $f(x) \cdot g(x) = 0$ for some $x \in S^n$, absurd.

Sps $f(x) \cdot g(x) > 0$ for all $x \in S^n$. Sps FTSOC $g(x) = -f(x)$ for some $x \in S^n$. Then $-f(x) \cdot f(x) = -1 > 0$, absurd. Hence, $g(x) \neq -f(x)$ for all $x \in S^n$, so $g \simeq f$, so $|\deg f| = |\deg g|$. Similarly, if $f(x) \cdot g(x) < 0$ for all $x \in S^n$, then $f \simeq ag$, so $|\deg f| = |\deg g|$, since $H_n(a) = (-1)^{n+1}$.

5 Spherical Complexes

Let M be a left R -module and $N \leq M$ be a submodule such that $N \cong R$ as R -modules. Then $N = \langle x_0 \rangle$ for some nonzero $x_0 \in M$.

Proof: Let $\varphi : R \rightarrow N$ be the isomorphism. Let $x_0 = \varphi(1)$. I claim that $N = \langle x_0 \rangle$ and that $x_0 \neq 0$. Since φ is an isomorphism, φ is also a set bijection. It is assumed $|R| > 1$, so $x_0 \neq 0$.

Suppose $y \in N$. Then $y = \varphi(r) = r\varphi(1) = rx_0$ for some $r \in R$. Conversely, if $r \in R$, then $rx_0 \in N$.

Suppose $f : R^2 \rightarrow R$ is an R -module homomorphism such that $f(1, -1) = 0$, $\ker f \cong R$, and $f(a, 0), f(0, b) \neq 0$ for all $a, b \neq 0$. Then $\ker f = \langle (1, -1) \rangle$.

Proof: $\ker f \leq R^2$ is a submodule. By the previous result, $\ker f = \langle (a, b) \rangle$ for a, b both not zero. Then $(a, b) - a(1, -1) = (0, a+b) \in \ker f$, so $a+b = 0 \rightarrow a = -b$. Hence, $(a, b) = a(1, -1)$.

6 Problems

Let $f, g : C \rightarrow C'$ be chain maps. Then $f + g : C \rightarrow C'$ is a chain map and $H(f + g) = H(f) + H(g)$.

Proof: We have $(f_q + g_q)(\partial_{q+1}x) = f_q\partial_{q+1}x + g_q\partial_{q+1}x = \partial_q f_q x + \partial_q g_q x = \partial_q(f_q + g_q)x$, and $f_q + g_q$ is a module homomorphism.

Let $z \in Z_q(C)$. Then $H_q(f+g)(\bar{z}) = \overline{(f+g)_q(z)} = \overline{f_q(z) + g_q(z)} = \overline{f_q(z)} + \overline{g_q(z)} = H_q(f)(\bar{z}) + H_q(g)(\bar{z})$.

Let $f, g : C \rightarrow C'$ be chain maps such that $C, H(C)$ are free. Suppose $H(f) = H(g)$. Then $f \simeq g$? $f - g : C \rightarrow C'$ is a chain map with $H(f - g) = H(f) - H(g) = 0$. Then $f - g \simeq 0$.

Claim: If $f_1 \simeq f_2$ and $g_1 \simeq g_2$, then $f_1 + g_1 \simeq f_2 + g_2$.

Proof: Let $D_q, E_q : C_q \rightarrow C'_{q+1}$ be the diagonal maps for $f_1 \simeq f_2, g_1 \simeq g_2$. I claim that $D_q + E_q$ is the diagonal map for $f_1 + g_1 \simeq f_2 + g_2$. Indeed,

$$\begin{aligned} & \partial'_{q+1}(D_q + E_q)x + (D_{q-1} + E_{q-1})\partial_q x \\ &= (\partial'_{q+1}D_q x + D_{q-1}\partial_q x) + (\partial'_{q+1}E_q x + E_{q-1}\partial_q x) \\ &= (f_{1q} - f_{2q}) + (g_{1q} - g_{2q}) = (f_{1q} + g_{1q}) - (f_{2q} + g_{2q}). \end{aligned}$$

Let $H(f) = 0, C, H(C)$ free. Consider the exact sequence

$$H_q(C) \xrightarrow{H_q(f)} H_q(C') \rightarrow H_q(Cf) \rightarrow H_{q-1}(C) \xrightarrow{H_{q-1}(f)} H_{q-1}(C'),$$

so we have the short split exact sequence

$$0 \rightarrow H_q(C') \rightarrow H_q(Cf) \rightarrow H_{q-1}(C) \rightarrow 0,$$

so $H_q(Cf) \cong H_q(C') \oplus H_{q-1}(C)$ as modules.

We wish for $D_q : C_q \rightarrow C'_{q+1}$ such that $\partial'_{q+1}D_q + D_{q-1}\partial_q = f_q$. For $q = 0$, we have $\partial'_1 D_0 = f_0$. For $x \in C_0$, we have $f_0(x) \in B_0(C')$ since $C_0 = Z_0(C)$.

Let R be a PID, C be a chain complex such that $C, H(C)$ are free. Let $f : C \rightarrow C'$ be a chain map such that $H(f) = 0$. Then $f \simeq 0$.

Proof: We have the exact sequence $0 \rightarrow Z_q \xrightarrow{i} C_q \xrightarrow{\partial} B_{q-1} \rightarrow 0$. Since C_{q-1} is free and R is a PID, so is $B_{q-1} \leq C_{q-1}$. The above sequence splits, so let $\pi_q : C_q \rightarrow Z_q$ be a projection. Further, we have the split exact sequence $0 \rightarrow B_q \xrightarrow{j} Z_q \xrightarrow{p} H_q \rightarrow 0$ since H_q is free. Let $\tau_q : Z_q \rightarrow B_q$ be a projection.

Denote by $E_q = B_q \oplus H_q \oplus B_{q-1}$. Define $\psi_q : C_q \rightarrow E_q$ by $\psi(x) = (\tau\pi x, p\pi x, \partial x)$. Define the operator ∂ on $E = \{E_q\}_{q \in \mathbb{Z}}$ by $\partial(x_1, x_2, x_3) = (x_3, 0, 0)$. Note that $\partial^2 = 0$ in E , so (E, ∂) is a chain complex. I claim that ψ is a chain isomorphism.

First, I claim ψ is a chain map. Let $x \in C_q$. Note that $\pi\partial x = \pi i(\partial x) = \partial x$. Similarly, $\tau\partial x = \partial x$. Then

$$\psi\partial x = (\tau\pi\partial x, p\pi\partial x, \partial\partial x) = (\tau\partial x, p\partial x, 0) = (\partial x, 0, 0)$$

and

$$\partial\psi x = \partial(\tau\pi x, p\pi x, \partial x) = (\partial x, 0, 0),$$

so ψ is indeed a chain map. From henceforth we work with E instead of C .

As modules, $\phi_1 : C_q \rightarrow Z_q \oplus B_{q-1}$ by $x \mapsto (\pi x, \partial x)$ is an isomorphism. Similarly, $\phi_2 : Z_q \rightarrow B_q \oplus H_q$ by $x \mapsto (\tau x, p x)$ is an isomorphism, so $\psi = (\phi_2 \oplus 1)\phi_1$ is a module isomorphism. Hence, ψ is a chain isomorphism.

Suppose $f : E \rightarrow C'$ has $H(f) = 0$. Note that $Z_q(E) = B_q \oplus H_q$, so $f(x_1, x_2, 0) \in B_q(C')$ for all $x_1 \in B_q, x_2 \in H_q$. Since H_q is free, there exists a module homomorphism $\varphi'_q : B_q \oplus H_q \rightarrow C'_{q+1}$ such that $f(0, x_2, 0) = \partial'_{q+1}\varphi'_q(x_1, x_2)$ and $\varphi'_q(x_1, 0) = 0$ for all $x_1 \in B_q$.

Let $\varphi_q : E_q \rightarrow C'_{q+1}$ be given by $\varphi_q(x_1, x_2) = \varphi'_q(x_1, x_2) + f_{q+1}(0, 0, x_1)$ and define $D_q : E_q \rightarrow C'_{q+1}$ by $D_q(x_1, x_2, x_3) = \varphi_q(x_1, x_2)$. I claim that D_q is our desired diagonal map. Indeed,

$$\begin{aligned} \partial'_{q+1}D_q(x_1, x_2, x_3) + D_{q-1}\partial_q(x_1, x_2, x_3) &= \partial'_{q+1}(\varphi'_q(x_1, x_2) + f_{q+1}(0, 0, x_1)) + D_{q-1}(x_3, 0, 0) \\ &= f_q(0, x_2, 0) + f_q(x_1, 0, 0) + \varphi_q(x_3, 0) = f_q(0, x_2, 0) + f_q(x_1, 0, 0) + \varphi'_q(x_3, 0) + f_q(0, 0, x_3) \\ &= f_q(x_1, x_2, x_3). \end{aligned}$$

Let M be a finitely generated module and $f : M \rightarrow M$ be a surjective endomorphism. Then f is an isomorphism.

Proof: Sps M is free with basis $\{e_i\}_{i \in [n]}$. Then $f = A$ for a matrix $A \in M_n(R)$. Since f is surjective, there exists $x_i \in M$ such that $Ax_i = e_i$, so $AB = I$ for some $B = [x_i]_{i \in [n]}$.

Lemma: Let G be a set with a binary operation $\cdot$ and identity 1 such that for all $g \in G$, we have $1g = g1 = g$, and there exists $h \in G$ such that $gh = 1$. Further, assume that $\cdot$ is associative. Then $hg = 1$, so that G is a group.

Proof: Let $x \in G$ such that $hx = 1$. Then $g = g1 = g(hx) = (gh)x = 1x = x$.

Let R be commutative. Denote by $M_n^\times(R) = \{A \in M_n(R) : \det A \in R^\times\}$. I claim that $M_n^\times(R)$ is a group.

Proof: We have $A \cdot \text{Adj}(A) = \det A \cdot I$. Then A has right inverse $\det^{-1} A \cdot \text{Adj}(A) \in M_n^\times(R)$, which is also the left inverse.

Now $AB = I \rightarrow \det A \cdot \det B = 1 \rightarrow \det A, \det B \in R^\times$, so $A, B \in M_n^\times(R)$, so $BA = I$. Let g denote the homomorphism with matrix B . Then $gf = fg = 1$, ie f is an isomorphism.

Claim: $S^n \times S^m$ is a CW complex for all $n, m \geq 1$.

Proof: We first have a lemma: Let $A \subseteq X$ and $B \subseteq Y$. Then $(A \times B)^\circ = A^\circ \times B^\circ$. Indeed, $A^\circ \times B^\circ$ is open and contained in $A \times B$. Let $U \subseteq A \times B$ be open. Then $\pi_1(U) \subseteq A$ and $\pi_1(U)$ is open, since π_1 is an open map. Then $\pi_1(U) \subseteq A^\circ$ and similarly $\pi_2(U) \subseteq B^\circ$, so $U \subseteq \pi_1(U) \times \pi_2(U) \subseteq A^\circ \times B^\circ$.

It follows as a subset of $\mathbb{R}^n \times \mathbb{R}^m$, we have

$$\begin{aligned} \partial(E^n \times E^m) &= E^n \times E^m - (E^n \times E^m)^\circ = E^n \times E^m - (E^n)^\circ \times (E^m)^\circ \\ &= [(E^n - (E^n)^\circ) \times E^m] \cup [E^n \times (E^m - (E^m)^\circ)] \\ &= (\partial E^n \times E^m) \cup (E^n \times \partial E^m). \end{aligned}$$

Let $f_n : (E^n, S^{n-1}) \rightarrow (S^n, x_0)$ be the usual map. Let $\phi : E^{n+m} \rightarrow E^n \times E^m$ be a homeomorphism. Note that $\phi((E^{n+m})^\circ) = \phi(E^{n+m})^\circ \rightarrow \phi((E^{n+m})^\circ) = (E^n)^\circ \times (E^m)^\circ$, so ϕ is an isomorphism from $\partial(E^{n+m})$ to $\partial(E^n \times E^m)$.

Note that

$$\begin{aligned} S^{n+m-1} &= \partial E^{n+m} \xrightarrow{\phi} \partial(E^n \times E^m) = (\partial E^n \times E^m) \cup (E^n \times \partial E^m) \\ &\xrightarrow{f_n \times f_m} (x_0 \times S^m) \cup (S^n \times y_0) = S^n \vee S^m \end{aligned}$$

and

$$\begin{aligned} (E^{n+m})^\circ &\xrightarrow{\phi} (E^n)^\circ \times (E^m)^\circ \xrightarrow{f_n \times f_m} (S^n - x_0) \cup (S^m - y_0) \\ &= S^n \times S^m - ((x_0 \times S^m) \cup (S^n \times y_0)) = S^n \times S^m - (S^n \vee S^m), \end{aligned}$$

where the arrows of the second sequence are isomorphisms, so $(f_n \times f_m)\phi : (E^{n+m}, S^{n+m-1}) \rightarrow (S^n \times S^m, S^n \vee S^m)$ gives the isomorphism $S^n \times S^m \cong (E^n \times E^m) \cup_\varphi (S^n \vee S^m)$, where $\varphi = (f_n \times f_m)\phi|_{S^{n+m-1}}$.

7 Excision Theorem Review

For $q \geq 0$ define $G_q = \frac{1}{q+1} \sum_{i=0}^q E_q$. Let $G_q \delta_q = (G_q, E_0, E_1, \dots, E_q)$ and $G_q \sigma = S_{q+1}(\sigma) G_q \delta_q$.

Define $\text{Sd}\delta_q = G_{q-1}\partial_q\delta_q$ and $T_q\delta_q = G_q(\delta_q - \text{Sd}\delta_q)$. Then we compute

$$\begin{aligned}
& \partial_{q+1}T_q\delta_q + T_{q-1}\partial_q\delta_q = \partial_{q+1}G_q(\delta_q - \text{Sd}\delta_q) + G_{q-1}(\partial_q\delta_q - \text{Sd}\partial_q\delta_q) \\
& = \partial_{q+1}G_q\delta_q - \partial_{q+1}G_qG_{q-1}\partial_q\delta_q + G_{q-1}\partial_q\delta_q - G_{q-1}G_{q-2}\partial_{q-1}\partial_q\delta_q \\
& = (\delta_q - G_{q-1}\partial_q\delta_q) - \partial_{q+1}G_qG_{q-1}\partial_q\delta_q + G_{q-1}\partial_q\delta_q \\
& = \delta_q - \partial_{q+1}G_qG_{q-1}\partial_q\delta_q \\
& = \delta_q - (G_{q-1}\partial_q\delta_q - G_{q-1}\partial_qG_{q-1}\partial_q\delta_q) \\
& = \delta_q - (G_{q-1}\partial_q\delta_q - G_{q-1}(\partial_q\delta_q - G_{q-2}\partial_{q-1}\partial_q\delta_q)) \\
& = \delta_q.
\end{aligned}$$